

EXECUTIVE PROFILE

Talented **Biologist (PhD)** and **AI Automation Entrepreneur** with over 5 years of international research, analytical, and technological leadership experience in South Africa, Hungary, and Germany. Expertise encompasses cardiac optogenetics, single-cell patch-clamp electrophysiology, quantitative photocycle modeling, and production-grade agentic AI workflows. Proven track record of publishing high-impact research (*Science Advances*, *Biophysical Journal*), leading cross-functional teams, and engineering AI-driven operating systems that optimize operational throughput by 30%+. Seeking to leverage interdisciplinary scientific rigor and tech innovation in senior/executive roles across Biotechnology, Pharmaceuticals, R&D, HealthTech, and Enterprise Solutions.

PROFESSIONAL & RESEARCH EXPERIENCE

Founder & Lead AI Solutions Architect

Scinovalities AI Automation Agency — Johannesburg, South Africa

Jan 2025 — Present

- Architected enterprise-grade autonomous voice agents and agentic AI workflows, reducing candidate screening processing times by 60% and accelerating talent acquisition.
- Engineered a scalable Hospitality Operating System, automating 40% of manual administrative tasks via modern technology orchestration.
- Developed 'Autoflow AI' (Real Estate & Lodging Management System), reducing repetitive manual workflows and elevating team project delivery speed by 25%.
- Created 'Lead Genius SA', an automated B2B lead generation platform integrated with CRM, multi-channel outreach, bulk email automation, and marketing analytics.
- Designed and optimized digital presence and brand assets for enterprise clients, achieving a +28% increase in operational throughput.

LLM Orchestration	Agentic AI	Voice Agent Deployment	Modern Technology	Scalable API Design
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Doctoral Research Scientist

Institute for Experimental Cardiovascular Medicine (IEKM), University of Freiburg — Germany

Nov 2021 — Mar 2026

- Investigated novel light-gated potassium channels for optical modulation and inhibition of cardiomyocyte action potentials and tissue excitability to advance anti-arrhythmic research.
- Orchestrated and executed high-throughput single-cell patch-clamp electrophysiology experiments using Axopatch 200B amplifiers and Digidata 1550A digitizers.
- Engineered adenoviral expression vectors and performed primary mammalian cardiomyocyte isolations under strict S2 Good Laboratory Practice (GLP) standards.
- Co-developed quantitative photocycle models for novel optogenetic channels, directly contributing to research published in *Science Advances* (2025) and *Biophysical Journal* (2026).
- Presented peer-reviewed scientific findings at premier international conferences across Europe, including NoTiCE (Glasgow), SPP1926 (Uslar), and DGK (Mannheim).
- Organized and led weekly IEKM PhD Journal Clubs (2022–2023), driving critical literature analysis and knowledge sharing among doctoral candidates.

Quantitative Modeling	Signal Processing	Patch-Clamp Electrophysiology	Data Mining	S2 GLP
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Master's Research Fellow

Department of Biochemistry, Eötvös Loránd University (ELTE) — Budapest, Hungary

Sep 2017 — Jul 2019

- Elucidated the biochemical mechanics of RecQ helicases in homologous recombination pathway selection to decipher fundamental genome stability mechanisms.
- Purified high-yield recombinant proteins and plasmid DNA, optimizing MaxPrep protocols and expression vector constructs.
- Quantified protein-DNA binding dynamics using advanced biophysical assays, including Fluorescence Anisotropy and Electrophoretic Mobility Shift Assays (EMSA).
- Evaluated molecular recombination pathways to characterize helicase-protein interactions within cell-free expression systems.

Biophysical Modeling	Statistical Analysis	Protein Purification	Fluorescence Anisotropy
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Graduate Research Assistant

Johannesburg Forensic Pathology Services / Wits University — South Africa

Feb 2015 — Nov 2016

- Evaluated toxicological medico-legal autopsy records (2013–2015) to identify mortality patterns and toxicological trends across Greater Johannesburg.
- Analyzed, sanitized, and categorized extensive laboratory case records, extracting critical toxicological data for statistical synthesis.
- Synthesized a comprehensive forensic thesis connecting toxic substance prevalence to medico-legal cause-of-death findings.

Data Analytics	Forensic Toxicology	Statistical Synthesis	Case Database Review
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Teaching Assistant & Demonstrator

Jun 2015 — Jul 2015

Targeting Talent Program, University of the Witwatersrand (Wits) — South Africa

- Delivered laboratory instruction and academic mentorship to top-performing Grade 12 STEM scholars.
- Demonstrated core chemistry experiments and molecular biology techniques to over 30 high-achieving scholars.

EDUCATION & ACADEMIC QUALIFICATIONS

PhD in Biology (Cardiac Optogenetics)

Nov 2021 — Mar 2026

University of Freiburg — Freiburg, Germany

- *Research Focus:* Light-gated potassium channels and optical modulation of cardiac rhythm.

MSc in Biology (Molecular, Cellular & Developmental Biology)

Sep 2017 — Jul 2019

Eötvös Loránd University (ELTE) — Budapest, Hungary

- *Research Focus:* Specialized in RecQ helicases and homologous recombination selection pathways.

BHSc (Honours) in Forensic Sciences

Feb 2015 — Nov 2015

University of the Witwatersrand (Wits) — Johannesburg, South Africa

- *Research Focus:* Forensic toxicology, medico-legal pathology, and molecular diagnostics.

BSc in Biological Sciences (Biochemistry, Cell Biology & Genetics)

Feb 2013 — Nov 2014

University of the Witwatersrand (Wits) — Johannesburg, South Africa

- *Research Focus:* Double major in Biochemistry & Cell Biology, and Genetics & Developmental Biology.

TECHNICAL SKILLS & CORE COMPETENCIES

AI & Automation: LLM Orchestration, Agentic Workflows, Autonomous Voice Agents, API Design, Modern Tech Stack, Linux.	Electrophysiology & Optics: Single-Cell Patch-Clamp (Axopatch 200B, Digidata 1550A), Langendorff Perfusion, Optogenetics.
Data Science & Analytics: Signal Processing, OriginPro 2024b, Clampfit, pClamp, IonOptix, Data Mining, Quantitative Modeling.	Molecular & Cell Biology: Adenoviral Vector Design, DNA/Protein Purification (MaxPrep), Cell Culture, EMSA, Anisotropy.
Forensic Pathology & Science: Forensic Toxicology, Medico-Legal Database Review, Forensic Genetics.	Leadership & Governance: Research Mentorship, PhD Journal Club Lead, Student Council Secretary, FELASA B, S2 GLP.

PUBLICATIONS & ACADEMIC CONFERENCES

- **Spreen et al. (2025).** Optogenetic silencing by combining a rhodopsin cyclase with an engineered cGMP-gated potassium channel. *Science Advances*, 11(48).
- **Ohnemus, Tillert, Leemisa et al. (2026).** Experimentally informed photocycle model of light-gated channel WiChR. *Biophysical Journal*, 1(17).
- **NoTiCE International Symposium (2023):** Researcher & Poster Presenter — Glasgow, Scotland, UK.
- **SPP1926 Next Generation Optogenetics Conference (2023):** Co-Author & Presenter — Uslar, Germany.

CERTIFICATIONS, FELLOWSHIPS & HONORS

FELASA Category B & S2 GLP Biosafety Certification University of Cologne, Germany (2021/2022)	IEKM Doctoral Research Fellowship University of Freiburg / IEKM (2021)
Good Scientific Practice & Advanced Biostatistics Spemann Graduate School of Biology & Medicine (2022)	Stipendium Hungaricum Master's Scholarship Hungarian Government & ELTE (2017)
Mental Health in Academia Credential Spemann Graduate School of Biology & Medicine (2023)	NRF Honours Funding & Wits Entrance Scholarship National Research Foundation & Wits Univ. (2013/2015)

PROFESSIONAL REFERENCES

Dr. Franziska Schneider-Warme Optogenetics Group Leader / PhD Supervisor IEKM, University of Freiburg, Germany franziska.schneider.uhz@uniklinik-freiburg.de	Prof. Dr. Med. Peter Kohl Director & PhD Co-Supervisor IEKM, University Heart Center Freiburg, Germany peter.kohl@uniklinik-freiburg.de
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